

Australian Math Olympiad 2016 — P5

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If a, b, c are positive integers such that $a^3 + b^3 = 2^c$, prove that $a = b$.

Solution

WLOG assume that $a \geq b$. Let $g = \gcd(a, b)$. For some coprime positive integers x, y we may write $a = gx$ and $b = gy$. We have

$$g^3(x^3 + y^3) = 2^c.$$

Since $(x^3 + y^3) \mid 2^c$, this quantity must be a pure power of 2. Since $x, y \in \mathbb{N}$ we have $x^3 + y^3 = 2^n$ for some $n \in \mathbb{N}$. Reducing mod 2 we find that x, y must have the same parity, but they cannot both be even since $\gcd(x, y) = 1$, so the only possibility is that x, y are both odd. Notice that

$$x^3 + y^3 = (x + y)(x^2 - xy + y^2) = 2^n.$$

Since both x, y are odd we observe that $x^2 - xy + y^2 \equiv 1 \pmod{2}$. Since $x^3 + y^3$ is a pure power of 2, the only possibility is that $x^2 - xy + y^2 = 1$. Now suppose FTSOC that $a > b$ or equivalently $x > y$. Then

$$1 = x^2 - xy + y^2 > (y + 1)^2 - (y + 1)y + y^2 = y^2 + y + 1 \geq 3$$

which is a contradiction. Thus $a \leq b$, but since we assumed also $a \geq b$, the only possibility is $a = b$. ■